MATH 257 Project I: Why use multivariate data analysis? - A Comparison between Univariate Confidence Intervals and Confidence Regions

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Introduction

The introduction briefly describes to the reader 1) the key research question or knowledge gap that is addressed by the study; 2) the data used for this study; 3) the methods used to investigate the research question; and 4) the key findings and why they are important. Three to four paragraphs should be sufficient.

Why do we use multivariate data analysis? When you have multiple response variables that aren't perfectly correlated, standard confidence intervals do not offer calibrated simultaneous coverage. Confidence regions do offer accurate coverage guarantees in multiple dimensions, but they are challenging to describe and use.

Recall the univariate t-based confidence interval for a population mean:

$$\bar{x} \pm t_{n-1}(\alpha/2)\frac{s}{\sqrt{n}}$$

This confidence interval rejects all values of μ_0 outside of its bounds and accepts all values of μ_0 inside its bounds, corresponding to the null hypothesis

$$H_0 : \mu = \mu_0$$

A confidence region, $R(\mathbf{X})$ ## Data This section should 1) describe the observational units of the dataset; 2) introduce all relevant variables; 3) describe any relevant details about how the data were generated or collected (and if necessary, transformed); and 4) highlight any critical characteristics of the data, possibly by using visualizations or summary tables

We generate the data matrix X

Methods

The paper should present the univariate and multivariate methods used in your analysis clearly using mathematical notation and plain English explanations. Every symbol and component of the models should be introduced and described. Your target reader should be a peer in this class. Focus on 1) clearly explaining the key distinctions between the univariate/multivariate methods; 2) explaining how you will compare the methods using your data; and 3) describing what you expect to see in your results (and why).

Results

The results section should describe the fitted models and the results of analysis. This section should include relevant tables, graphs, and quantitative results, as well as plain English explanations of all of these components, but should not include raw R output. Clearly explain what the results show and interpret them in the context of your research question

Discussion

In the final section of the paper, briefly summarize your key findings, discuss potential weaknesses of your study, and identify potential improvements and extensions or future areas of related research.

References

Quarto enables you to weave together content and executable code into a finished document. To learn more about Quarto see https://quarto.org.

Running Code

```
library(MASS)
n = 1000
mu = c(x1=5,x2=10,x3=-2)
Sigma <- matrix(c(1.0, 0.5, 0.2,
                  0.5, 2.0, 0.8,
                  0.2, 0.8, 1.5),
                nrow = 3, ncol = 3)

# Set seed for reproducibility
set.seed(0)
```

```r
# 4. Generate the multivariate normal data
simulated_data <- mvrnorm(n = n, mu = mu, Sigma = Sigma)

# View the first few rows
head(simulated_data)
```

```
           x1        x2         x3
[1,] 4.857378  8.244567 -3.2273246
[2,] 4.086633  9.838151 -0.3090197
[3,] 4.451708  8.326446 -3.3331632
[4,] 5.007781  8.924934 -4.3042173
[5,] 5.986992  9.697969 -3.3346212
[6,] 5.379122 12.300702 -0.8392455
```

```r
# Bivariate normal data
library(mvtnorm)
```

```
Warning: package 'mvtnorm' was built under R version 4.5.3
```

```r
mu = c(x1=0,x2=0)
Sigma = matrix(c(1, .33,
                 .33, 1), nrow=2)
n=30
data = rmvnorm(n=n,mean=mu,sigma=Sigma) |>
  as.data.frame()

data
```

```
            x1          x2
1  -0.44736451  0.39953993
2  -1.60783398 -1.33823808
3  -0.90528544  0.38956601
4  -0.28781827 -0.48815950
5   2.12841310  1.66258910
6   0.22731571 -1.16333477
7  -0.38401611 -2.36184312
8   0.17472114  1.33511828
9  -0.38484254 -0.50614149
10  0.68111132  1.79852527
11 -0.70620440 -0.31634699
```

```
12  0.25693664  0.57499739
13  0.32594240 -0.31637257
14 -0.08666672  0.13240913
15  1.51515705  1.24386862
16 -0.82212442 -0.24023145
17 -1.69876752 -0.09443339
18  0.77305550  0.56352825
19  0.55082053  0.59260784
20  1.06934872  0.74912346
21 -1.45254837 -0.85910577
22  0.13092036  1.10486304
23 -0.28593075  0.02000762
24  0.04105518 -0.66945049
25  0.91365951  0.67581692
26  0.48036445  0.36820249
27 -1.57708429 -1.66325319
28  0.05398109  0.56044992
29  0.78757662 -0.21469328
30  0.35660006 -0.62746019
```

```
confint_x1 = t.test(data$x1)$conf.int
confint_x1
```

```
[1] -0.3467543  0.3347871
attr(,"conf.level")
[1] 0.95
```

s